CresCine's Film Industry Data Repository (FIDA)

Indrek Ibrus[1], Manuel Pita[2], Gabriela Soares[2], João Pedro Carvalho[2], Thiago Paiva[2], Ksenia Mukhina[1], Andres Karjus[1], Ana Falcón[1], Zuil Pirola[2], Bruno Saraiva[2], Marius Øfsti[3], Aksel Nõmmela[1], Alisa Zeleva[4], Jukka Huhtala[4], Jenny Grouiller-Ruhland[4], Saara Hyvönen[4], Timo Norros[4], Sergei Posysaev[4], Vejune Zemaityte[5]

[1] Tallinn University, Baltic Film, Media and Arts School

[2] Lusofona University, The Centre for Research in Applied Communication, Culture, and New Technologies

[3] Aarhus University, Department of Media and Journalism Studies

[4] DAIN Studios

[5] independent research

Corresponding author: Indrek Ibrus, professor of media innovation, Tallinn University, Estonia; indrek.ibrus@tlu.ee

Acknowledgements: Production of FIDA and the writing of this article have been supported by the European Union (grant nr 101094988).

Abstract

The Film Industry Data Repository (FIDA) is a lifecycle-wide, multi-source database developed by the CresCine consortium[1] to address the persistent data scarcity facing Europe's small and mid-sized film markets. Built on a scalable Databricks architecture and structured through a medallion pipeline (Bronze–Silver–Gold), FIDA integrates heterogeneous datasets covering production metadata, festival circulation, theatrical distribution (showtimes, admissions, box-office), streaming availability, television programming, and socio-economic context. Data from public and open infrastructures (TMDB, Wikidata, Lumiere, World Bank), institutional partners (Cinando, European Audiovisual Observatory), and selected commercial providers (International Showtimes, UsherU, media-press.tv) are cleaned, harmonised, and linked through an internal

[1] https://www.crescine.eu/

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Executive Agency. Neither the European Union nor the European Research Executive Agency can be held responsible for them.

identifier (CresCine ID) using deterministic and fuzzy-matching techniques. The resulting star-schema repository enables cross-window, cross-territory analysis of European films with a granularity not previously available, especially for countries underrepresented in commercial analytics services. FIDA is disseminated through interactive analytical dashboards and simulation tools, supported by the release of specialised aggregated datasets that comply with licensing restrictions. Designed for long-term sustainability and interoperability, FIDA provides a durable evidence base for researchers, policymakers, and industry stakeholders seeking to understand and strengthen European film circulation, performance, and public value creation.

Keywords

Film production, film distribution, film data, film analytics, European film market

Background

The European film industry operates through a multifaceted value chain that spans production, festival circulation, theatrical exhibition, and subsequent distribution windows. Yet the data generated across these stages is collected unevenly, stored in heterogeneous formats, and seldom interlinked in ways that enable systematic comparison across windows, territories, or time periods. Existing commercial data services that do integrate multi-window information typically prioritise the interests of large film markets, offering limited visibility into smaller European countries such as Estonia, Lithuania, Croatia, Portugal, Denmark, Ireland, or Belgium. As a result, researchers as well as public and private stakeholders in these countries face significant barriers to accessing comprehensive, high-resolution information about their domestic industries or related markets.

CresCine—a pan-European research consortium dedicated to strengthening the resilience of Europe's film ecosystem—addresses this structural gap through the development of the Film Industry Data Repository (FIDA). FIDA aggregates structured and unstructured datasets from across the European film domain, encompassing production metadata, festival screenings and awards, theatrical performance, streaming availability, and television showtimes.

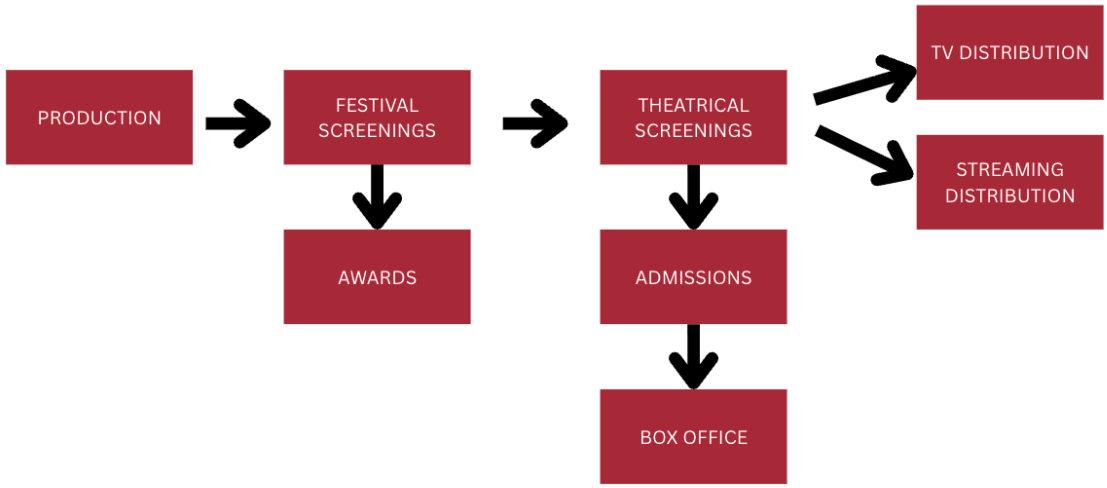


Figure 1: Simplified lifecycle of a film from production to distribution.

This image depicts how a film produced in Europe is typically commercialised. While there are many different ways in which films can complete each stage that the one captured in this figure, this general overview can help to visualise the stages in which data is produced.

Figure 1 offers a simplified depiction of the film lifecycle and the moments at which key data is generated. A film first enters production, generating metadata such as genre, production countries, spoken languages, and runtime. Upon completion, it circulates through festivals, where initial screenings and industry recognition produce further data points. The subsequent theatrical cycle—first domestic, and later international—creates

country-level information on admissions and box-office revenues. After the theatrical window closes, films continue to circulate on broadcast television and video-on-demand platforms, producing additional layers of distribution and availability data. FIDA integrates data from all these stages, enabling lifecycle-wide analyses of how films from different European countries circulate and perform across markets and media environments.

The potential of FIDA is exemplified through two kinds of dashboards: first, analytics tools to visualise relationships across the film life cycle and, second, a theatrical distribution simulation tool. Together, these applications support the development of knowledge and strategies among small European film stakeholders, enhancing their competitiveness.

The design of FIDA is directly shaped by the persistent information gaps affecting Europe's smaller film markets. Commercial analytics firms—such and other global data providers—tend to focus their data-collection efforts on large territories where commercial returns justify detailed monitoring. As a result, stakeholders in smaller markets often lack access to high-resolution data on their own domestic film industries, making comparative analysis across windows and territories difficult. FIDA addresses this gap by integrating data sources that offer universal or near-universal coverage, such as TMDB, Wikidata, International Showtimes, and Usheru, alongside specialised European sources like Lumiere Pro. These datasets collectively provide granular information for all markets rather than privileging only the largest territories. By linking such sources into a unified lifecycle-wide architecture, FIDA enables analyses of production, festival circulation, theatrical distribution, streaming availability, and television programming even for countries that are otherwise invisible in commercial data systems. This design ensures that small-market stakeholders can access evidence that is comparable in quality and granularity to what is typically available only for large film economies.

The creation of FIDA is focused on technical interoperability, making disparate datasets to work together. Due to the lack of shared identifiers between the data sources, this work has meant various

challenges and workarounds. In the next pages, the architecture of FIDA is explained, presenting its data sources, data set contents, methods, schema, tables, and usage regulations.

Data sources used

The CresCine consortium undertook a systematic effort to collect data from multiple sources covering all stages of the film value chain. In some areas, this effort was fully successful, while in others it remains partial. Below, we present an overview of the datasets that had been ingested and integrated into the database by November 2025. Because several datasets originated from privately owned sources, it was necessary to negotiate the conditions under which such data could be made accessible. The resulting agreement stipulates that while FIDA may enable the discovery of analytic insights, it may not permit the publication of data relating to the production or distribution of individual films. Hence, all data is published in aggregate modes. All FIDA data sources are listed below, and the specific features of the data processing applied to each source are detailed in Annex 1.

A central design principle of FIDA was the long-term sustainability of the repository, which required selecting data sources that are legally accessible, financially viable, and technically suitable for integration into an open scientific infrastructure. Consequently, the CresCine consortium prioritised publicly accessible and non-proprietary datasets such as Lumiere and TMDB, as these enable persistent availability to European stakeholders without imposing restrictive licensing conditions. TMDB was preferred over IMDb specifically because the latter is proprietary, expensive to license at scale, and incompatible with the long-term open-access ambitions of the repository. In addition, FIDA incorporates data from Cinando and Lumiere Pro because these sources were made available to the project through institutional partnerships at no additional cost. Where essential data were not available through public channels, the consortium negotiated access to private commercial providers—most notably International Showtimes, Usheru, and media-press.tv—whose

terms and pricing structures permitted sustainable inclusion in FIDA. Several potentially valuable datasets could not be obtained despite negotiation efforts, including piracy datasets (e.g., MUSO), audience-level data from major streaming platforms, and reception-oriented data from Rotten Tomatoes, Letterboxd, and festival management systems such as Eventival. These limitations reflect structural barriers to data access in the film and streaming industries, and they have shaped the contours of FIDA's present data architecture.

The Movie Database (TMDB)

The TMDB dataset is a crowd-sourced repository of film metadata, with information such as title, cast, crew, genre, production companies, production countries, release dates, spoken languages, translations, production dates, release dates by country, and alternative titles.

FIDA features TMDB data from film productions released within the period 2013-2024. Alongside Lumiere Pro, TMDB provides the backbone of basic movie information for FIDA, which is then enriched with information from other ingested datasets.

Wikidata

The Wikidata-based dataset provides a rich, structured knowledge graph of films and their creative contributors. It captures extensive metadata, including creative personnel, awards, reviews, filming and narrative locations, box office performance, and languages.

FIDA includes Wikidata information from 2017 to 2024. WikiData information facilitates the multidimensional exploration of cinema, linking geographical, periodical, and production-related information with standardised identifiers from a variety of sources.

World Bank

The World Bank database provides socio-economic context for film market and audience studies. It contains demographic indicators such as population figures across countries and years, which can be

integrated with cinema-specific data to normalise performance metrics. FIDA includes World Bank data from the years 1990, 2000, and 2014 to 2023. The inclusion of World Bank data enables per-capita analyses of cinema attendance, film market penetration, and broader cultural consumption trends at the national and regional levels.

Cinando

Cinando is a proprietary online platform created in 2003 by the Marché du Film of the Cannes Film Festival. Initially a database of Cannes market attendees, it expanded into a year-round infrastructure which aids industry professionals navigate films, projects, contacts, and rights deals more efficiently. Beyond targeting individual users (such as producers, sales agents, distributors, buyers, festival programmers and sales companies), Cinando also serves institutional users: film festivals and film markets use it to schedule market screenings, host B2B screeners securely, coordinate rights sales, manage investment meetings, and access detailed information on films and professionals linked to their events. Its underlying relational database stores metadata on companies, people, films in development, screenings, market events, runtimes, production years, origin countries, languages, crew lists and content or "kind" tags (genre, target audience, identity labels, production type). For festivals specifically, Cinando records festival and event titles, location countries, event years and the full programming of films across thousands of events, enabling the reconstruction of film–festival co-occurrence networks and the analysis of programming diversity, hierarchy, and public value creation across more than 600 festival series worldwide (see Zemaityte et al., 2024). FIDA includes Cinando data covering festivals from 2012 to 2023. This data enables FIDA to study the screenings of European films in festivals across the world.

CresCine Festival Data

The CresCine Festival Data captures the programming, nominations and awards from the international festivals accredited by the International Federation of Film Producers Associations (FIAPF), as well as the

remaining festivals that have been included in the Swedish Film Institute (SFI) Quality Index. The combination of these two lists is understood to capture the most influential film festivals across the world.

Each record captures key festival-level attributes such as the festival name, edition, section, and award category, alongside jury status and ecosystem context. Film-level metadata is also included, covering both English and original titles, production year, production country, and associated directors. FIDA features CresCine Festival data for the period 2014-2023. This allows for the examination of festival selections, award patterns, thus providing a bridge between artistic recognition and broader film market data.

UsherU

Usheru is an Irish data-intelligence and content discovery service provider for the film and entertainment industries. It works primarily with institutional partners such as national film bodies, film institutes, studios and distributors, helping them to build direct-to-consumer relationships, track where films are available in cinemas and on streaming platforms, and optimise marketing spend. The core platform aggregates and standardises data on films, showtimes, streaming availability, campaigns and audience interactions. CresCine purchased from UsherU data about the availability of European (member countries of the Council of Europe) on streaming services across the world from the years 2021 to 2024 on a monthly basis. It includes information such as title, genre, release year, type of platform, streaming service, and dates of catalogue addition and catalogue removal.

Lumiere Pro

Lumiere Pro is the subscription-level offering from the European Audiovisual Observatory designed for film agencies, government bodies and institutional users who require deeper and more comprehensive theatrical market intelligence across Europe and key non-European territories. It builds on the publicly accessible LUMIERE admissions database but adds title-by-title box-office revenue and admissions data from a global measurement vendor (Comscore) for non-European markets as well as supplementary data sets for

European markets. The data covers annual admissions and gross box-office figures for individual feature films in each participating market, enabling public film bodies to analyse export patterns, market shares, temporal trends and performance of national catalogues versus foreign films. Institutional users thus gain access to a relational database infrastructure that supports statistical aggregation, visualisation and benchmarking of theatrical film circulation and performance across more than 50 markets worldwide (European and non-European). FIDA includes Lumiere Pro data with regard to the admissions and box office data of European films from the years 1996 to 2023.

International Showtimes

International Showtimes is a Berlin-based entertainment data company that is specialised in aggregating cinema and showtime information at a global scale. It operates a unified API that serves app developers, media and tech companies, marketing agencies and other institutional clients who need standardised data on which films are playing, where and at what time, across more than one hundred markets worldwide. The service ingests and normalises data from thousands of cinemas and ticketing partners, providing structured metadata on films, cinemas (locations, facilities) and individual showtimes. This infrastructure allows its clients to power film-discovery interfaces, "where to watch" services, campaign tracking dashboards and cross-border market analyses without building their own global data-collection systems, effectively turning showtimes and cinema programming into a machine-readable layer that other discovery, marketing and analytics tools can build upon. CresCine purchased for FIDA its showtimes data from the years 2021-2024. Since 2021-2023 data were raw JSON files that needed extensive cleaning efforts, and hence, not all data about all countries and years are currently included in FIDA.

European Audiovisual Observatory Yearbook

The European Audiovisual Observatory Yearbook is an online service which provides data on television, cinema, streaming and home video in European Countries. FIDA uses the EAO Yearbook as the source for

ticket price information. The data featured in FIDA covers price information from multiple international markets spanning the years 2014 to 2023 (historical data). This information enables the advanced analysis of cinema consumption as it enriches admissions, showtimes and box-office performance.

Media-press.tv

Media-press.tv is a European entertainment metadata company that specialises in collecting, structuring and selling detailed data about digital TV schedules and programmes. It aggregates electronic programme guide (EPG) and related metadata for hundreds of broadcasters and platforms, then licenses this data to TV operators, IPTV and cable providers, VOD and streaming services, magazine publishers and other media or internet service providers who need reliable listings and programme information. Its databases model linear and on-demand schedules at channel and programme level, including titles, synopses, genres, series–episode relationships, cast and crew, technical attributes and, for enhanced services, enriched sports and entertainment metadata with structured information about events, teams, venues and visuals. By delivering this data through standardised feeds and tooling, media-press.tv underpins a wide range of commercial EPGs, recommendation engines and TV guides, effectively turning TV schedules and programme line-ups into a machine-readable asset that others use to build user interfaces, search and discovery features and analytics around contemporary television. CresCine purchased for FIDA 4 years (2021-2024) of TV showtimes data - when and how European films have featured across 8000 European TV-channels. This enables to study the interrelationships of TV showtimes in relation to other release windows.

FIDA Dataset Contents

FIDA is the result of a three-layered process in which data was cleaned, matched and aggregated into what is called a Gold Layer. The layer follows a traditional star schema in which a core table (dim_filmmaster) connects to the other tables of the repository.

Table 1: Data tables of the Film Industry Data Repository (FIDA)

Table name	Type	Description
dim_filmmaster	Film Master Table	Film Production Information
dim_countryclassification	Metrics	Country classification
dim_genre	Dimension	All genres attached to a single film
dim_main_genres	Dimension	Main genre attached to a single film
dim_productioncountries	Dimension	Production Country
dim_spokenlanguages	Dimension	Spoken Languages
fact_admissions	Metrics	Admissions data
fact_tvshowtimes	Metrics	TV screening data.
fact_boxoffice	Metrics	Box Office data
fact_distributions	Metrics	VOD data
fact_market_results	Metrics	Screenings data

fact_festivals	Dimension	Festival Information
fact_ticketprices	Metrics	Ticket Prices

FIDA's principal strength lies in its capacity to combine heterogeneous data sources that represent different stages of the film industry value chain, yet this integrative ambition has also constituted its major technical challenge. Because the contributing databases were developed independently and rely on divergent identifier systems – including IMDb, TMDB, EIDR and ISAN – their records cannot be straightforwardly aligned. Considerable work has therefore been required to interlink and reconcile these sources, including the use of fuzzy-matching techniques to approximate correspondences where unique identifiers are absent or incomplete. Furthermore, certain APIs had restrictions on how data could be retrieved and navigated, which elongated the time dedicated to clean, prepare and aggregate data as described in the next section. This has meant that not all films or data points could be matched with full reliability, and some gaps in the unified master data table inevitably remain. Nonetheless, the architecture that has been developed enables robust analysis of film life cycles across multiple release windows and geographical markets, something that no single source previously allowed. Although FIDA cannot be made fully open due to licensing restrictions from commercial providers – which prohibit the publication of film-level records – the project mitigates this constraint by releasing a suite of analytical dashboards and aggregated datasets. These outputs make it possible for policymakers, researchers and industry stakeholders to examine the circulation, performance and reach of European films, including those from small and under-represented markets, through forms of integrated analysis that were not previously feasible. The actions taken to build FIDA are further discussed in the Methods Applied section and Annex 1.

\

Methods Applied

Different single film identifiers were present in each data source, with not a single one being repeated in all sources. In FIDA, these sources are joined through a comprehensive matchmaking process and the use of an internal ID known as CresCine ID. In addition, the creation of FIDA required processing the data in three stages, which resulted in the creation of three layers: Bronze, Silver, and Gold.

Data processing

One of the principles of FIDA is that data remains traceable, interoperable, reproducible, and updatable. Therefore, the databases used to build FIDA were processed through a three-layered approach based on the medallion architecture. Each layer (Bronze, Silver, Gold) was employed to fulfill a role within the data transformation pipeline spanning from raw data ingestion to the release of the refined datasets used to build the two FIDA industry dashboards.

The resulting layers support data updates as well as data pool extensions without disrupting its internal structure.

Bronze Layer (Raw Data Ingestion)

During this stage, data was ingested and saved in its near-original form in order to maintain schema consistency with future data updates. API-sourced data, CSV exports, and JSON files were stored in Delta format, in order to guarantee that any data point can be traced back to its original source. While data transformation was minimal, every row was enriched with a rowhash (to track versioning) and a metadata timestamp (to log when data was loaded and updated).

Silver Layer (Transformation and Integration)

Raw data was refined in the Silver Layer (SL) in order to improve interoperability between the FIDA datasets, enable the future integration of more data, and prepare the sources for integration in the Gold

Layer. Some of the key actions performed in this stage included assigning an internal ID (CresCine ID), cleaning and wrangling data, resolving data inconsistencies and performing other Quality Control (QC) measures, which are overviewed in the next section and detailed in Annex 1.

Gold Layer

The FIDA Gold Layer aims to support the creation of dashboards, data analyses, and simulation tools. Therefore, the layer follows a star schema, with a Film Master Table (dim_filmmaster) linking to multiple fact and dimension tables as detailed in the FIDA Dataset Contents section. In compliance with EU data protection regulations and to promote a fair usage of this public repository, identifiable data points are not included. The FIDA Gold Layer supports cross-domain analyses and data visualisation to study the lifecycle of films from world premiere to on-demand or TV distribution.

With its three-layered approach, FIDA is a data repository that can be scaled, updated, and extended. Therefore, FIDA is a database that approaches film analytics with a deep commitment to transparency and long-term sustainability. The details regarding the processing of each data source across the three FIDA layers are described in Annex 1.

Quality Control

Due to the diversity of data sources, Quality Control (QC) was a priority during the integration of FIDA in order to ensure that the data pool was consistent, interoperable, reliable and updatable. QC processes were implemented iteratively during raw data ingestion and transformation. Some of the QC controls performed included:

- Structural validation: encompassing completeness and coherence checks plus the identification of malformed records, invalid data points, and/or null values,
- ID Standardization and enrichment: each dataset was normalized through the creation of an internal identifier (*CresCine ID*) that allowed matching between different IDs (e.g., TMDB ID, Lumiere Pro

ID). Where identifiers were missing, fuzzy string matching was applied using field combinations such as the triplet film title x production year x production countries to identify as many titles as possible.

- Semantic standardization: fields such as genres, production countries and spoken languages were standardized. In addition, the contents of genres and spoken languages were also grouped into broader categories.
- Cross-source validation: Data points were cross-checked across sources for better accuracy.
- Data prep for traceability and version control: metadata fields identifying data source, loading data, and date of last update were added to identify the origin of each record. The FIDA Databricks environment is built for full versioning, so every transformation between the Bronze and Gold Layers can be audited and revised if necessary.

Through these Quality Control actions, FIDA achieves structural and semantic consistency. Thus the FIDA repository can serve both as a reliable corpus for research and as an interconnectable data framework that can be expanded.

Schema and tables

As mentioned before, the architecture of FIDA is a star schema, with the Film Master Table (dim_filmmaster) being a central database which connects to other repository components. Below is a short description of each of the tables contained within the FIDA Gold Layer.

Table 2: dim_filmmaster

The Master Table (*dim_filmmaster*) is the data backbone of FIDA. This table contains basic film data (e.g., production year, main production country, etc.) as well as the IDs necessary to connect its data to the other FIDA tables to create analyses and visualizations. To facilitate data searches, table joins, and filtering by fields, this table works independently.

Column	Type	Description
crescine_id	bigint	CresCine ID
tmdb_id	int	TMDB ID
lumiere_id	double	Lumiere Pro ID
title_original	string	Original film title
title_english	string	Film title in English
production_year	double	Year of Production
production_country_main	string	Main Production Country
release_date	string	Date of First Release
wiki_id	string	WikiData ID
imdb_id	string	IMDB ID
budget_in_eur	bigint	Budget in euros
budget_classification	string	Classification of budget

Table 3: dim_country

This table identifies the countries listed as production countries within FIDA, their total population, as well as their classification according to variables such as EU membership status.

Column	Type	Description
iso_code	string	Country ISO code
name	string	Name of country in English
population	string	Number of inhabitants of a

		country
council_of_europe_member	boolean	Status as European Union member
crescine	boolean	Status as CresCine focus country
council_of_europe_member	boolean	Status as a member of the European Union.
eu_small_countries	boolean	Status as a small country within the European Union
europe_small_countries	boolean	Status as a small country within Europe (EU and non-EU inclusive)
eu_big_5	boolean	Status as a big country within the European Union
europe_big_5	boolean	Status as a big country within Europe (EU and non-EU inclusive)

Table 4: dim_genre

This table identifies all the genres that can be attached to a film production.The table is the result of aggregating the genres field of Lumiere Pro, TMDB, and WIkidata.

Across data sources, a single production could be linked to multiple genres. In some data sources, films were categorized based on narrative style (e.g., sci-fi, horror, romance, etc.) while in others genre was used to refer to the type of film production (e.g., animation, live-action, documentary, etc.), running time (e.g., short, feature, etc.), and other discrete categories (e.g., first feature, student film, etc.).

Column	Type	Description
crescine_id	bigint	CresCine ID number

genres	string	All genres attached to a single film

Table 5: dim_main_genre

This table identifies main genres in which film productions can be classified. During the processing phase, the aggregated list of genres found across databases was sorted using Databricks Genie Large Language Model (LLM) in order to create a list of main genres that could contain all the genres present across data sources.

Column	Type	Description
crescine_id	bigint	CresCine ID
main_genre	string	Main genre attached to a single film

Table 6: dim_productioncountries

This table identifies the production country of each film. The production countries are classified to identify them as main production countries, and minority co-production countries.

Column	Type	Description
crescine_id	bigint	CresCine ID
production_country	string	Production Country
order	int	Distinction between main production country (order = 1) and coproduction countries (order = $[2,\infty)$)

number_of_production_countries	bigint	Number of production country

Table 7: dim_spoken_languages

This table identifies the spoken language(s) of every film listed in FIDA. The aggregated list of languages was sorted using LLMs, thus creating language families (e.g., French) and their associated languages (e.g., Standard French, Canadian French, Belgian French, etc.).

Since many productions feature more than one spoken language, the field *pos* is used to rank them by their prominence within a film. The value *0* identifies the main language of a production, while successive numbers list other languages spoken within a film.

Column	Type	Description
crescine_id	bigint	CresCine ID
language_family	string	Language Family (e.g., English, Spanish, French, etc.) without specifying a particular variant
sub_language	string	Language, specifying a particular variant (e.g., American English, Mexican Spanish, Belgian French, etc.)
pos	int	Order of spoken language, with the main one being 0
source	string	Source from which the spoken language data was retrieved (e.g., WikiData, TMDB, Lumiere Pro)

Table 8: fact_admissions

This table counts the number of theatrical admissions for each film production, organized by year and country in which the tickets were sold. To facilitate the analysis of domestic film consumption, films produced (either in a majority or minority capacity) within the studied country are also identified.

Column	Type	Description
crescine_id	bigint	CresCine ID
year	int	Year in which the admissions took place
market	string	Country ISO code
national	boolean	Indicates whether the film is a majority domestic production. The value is true if the film was primarily produced domestically, and false if this is not the case
national_extended	boolean	Indicates whether the film is a minority domestic production. The value is true if the film was partly (but not primarily) produced domestically, and false if this is not the case
admissions	int	Number of Admissions.

Table 9: fact_box_office

This table presents the box office in euros (€) achieved by each film, organized by year and country in which the revenue was generated. To ease the analysis of the performance of domestic productions, films produced (either in a majority or minority capacity) within the studied country are also identified.

Column	Type	Description
crescine_id	bigint	CresCine ID
year	int	Year in which the box office was recorded.
market	string	Country ISO code
national	boolean	Indicates whether the film is a majority domestic production. The value is true if the film was primarily produced domestically, and false if this is not the case.
national_extended	boolean	Indicates whether the film is a minority domestic production. The value is true if the film was partly (but not primarily) produced domestically, and false if this is not the case.
box_office_eur	int	Box Office in Euros (€)

Table 10: fact_distributions

This table features the distribution details for each production. Since films can have more than one first date of release (e.g., world premiere, festival debut, national release, etc.), additional information was included to facilitate the study of the film distribution cycle. A film can be released in more ways than ever before, with some movies being premiered in Video On Demand (VOD) platforms and others doing a theatrical release first. Thus, this table also contains information regarding the type of distribution, as well as the name of the distribution service used in the case of online releases.

Column	Type	Description
crescine_id	bigint	CresCine ID
release_date	date	Release date by distributor
first_release_date	date	First release date known (regardless of distributor)
first_theatrical_release_date	date	First theatrical release date known
market	string	Country ISO code
national	boolean	Indicates whether the film is a majority domestic production. The value is true if the film was primarily produced domestically, and false if this is not the case
national_extended	boolean	Indicates whether the film is a minority domestic production. The value is true if the film was partly (but not primarily) produced domestically, and false if this is not the case
distribution_type	string	Type of distribution (e.g., Theatrical, TV, SVOD, etc.)
distributor	string	Name of distribution platform, if any (e.g. Amazon Prime, Netflix, Google Play, etc.)
release_order_distribution_type	int	release order of distribution type for film (regardless in which market)
release_order_market_distributi	int	release order of distribution type

on_type		for film within a market
release_order_market	int	release order (of any distribution type) of markets (or market/country groups)

Table 11: fact_market_results

This table features the performance of each film within a specific market, including admissions and box office in euros (€). The table joins this information with the aim to support analyses related to ticket prices and revenue per capita. To ease the study of domestic films, the table also identifies films produced (either in a majority or minority capacity) within the studied country.

Column	Type	Description
crescine_id	bigint	CresCine ID
year	int	Year in which admissions were recorded
market	string	Country ISO code
national	boolean	Indicates whether the film is a majority domestic production. The value is true if the film was primarily produced domestically, and false if this is not the case
national_extended	boolean	Indicates whether the film is a minority domestic production. The value is true if the film was partly (but not primarily) produced domestically, and false if this is not the case

admissions	int	Number of admissions
showings	bigint	Number of screenings
box_office_eur	int	Box Office in Euros (€)

Table 12: fact_showings

This table features information regarding the number cinemas in which a film was screened on a particular date. Additional information was included to facilitate the study of theatrical screenings, since cinemas may have more than one screen and/or have several showtimes per screen.

Column	Type	Description
id	string	Record ID
date	date	Date of screening
cinema_id	int	Cinema ID
cinema_country	string	Country ID
is_id	int	International Showtimes ID
title	string	Film Title
crescine_id	bigint	CresCine FIlm ID

Usage Limitations

Scope of Use

The Film Industry Data Repository (FIDA) aggregates data on European film industries and markets, integrating datasets from public and private sources. Users may access, analyse, and visualise data through

the FIDA Analytics and the FIDA Simulation dashboards as well as by downloading the associated freely accessible datasets subject to the conditions described below.

Permitted Uses

- Users may employ FIDA for research, education, policy development, and industry strategy purposes.
- Users may generate analytical outputs, visualisations, and reports, provided proper attribution is given. Proper attribution is referencing this paper.
- Any reuse of FIDA data must comply with the following ****data source license <INSERT DATA LICENSE>

Restrictions

- Users may not attempt to re-identify anonymised/aggregated data.
- Users may not redistribute FIDA datasets, nor use FIDA to provide paid data services.
- Any simulation or analytical outputs remain the responsibility of the user; FIDA provides no warranty of accuracy.

Attribution

When using FIDA data or tools, users must refer to the repository by referring to this paper as follows:

"CresCine's Film Industry Data Repository (FIDA), CresCine, [Year of Access], DOI."

Liability

Users assume full responsibility for their analyses, outputs, and interpretations. FIDA authors and the CresCine consortium do not guarantee data completeness, accuracy, or fitness for specific queries.

Termination

Any violation of this license, including misuse of restricted data or data re-identification, may result in immediate suspension of access and possible legal action.

Reuse of the Data

FIDA data may be re-used for research, education, policy development, and industry strategy purposes. Users are allowed to write analyses, visualisations, and reports, provided that they include proper attribution. Re-use must comply with the license conditions described above.

Governance and Sustainability

The long-term sustainability of FIDA is ensured through a governance model anchored in the CresCine consortium during the project period and jointly maintained thereafter by Tallinn University and Lusófona University—the two leading academic institutions responsible for the repository's technical development. During the 2023–2026 project cycle, FIDA is being populated with data covering all major release windows across multiple years, enabling extensive experimentation with cross-market and cross-window analyses. After the formal conclusion of CresCine in March 2026, Tallinn University and Lusófona University will continue to manage, update, and expand the repository. This includes integrating new annual data from public and open sources, negotiating continued access to selected commercial datasets, and maintaining FIDA's analytical dashboards as the primary mode of public dissemination. The consortium is also exploring options for establishing a stable institutional home for FIDA as a long-term European film-data infrastructure. Updates to the repository will occur regularly, subject to the availability of new data from source providers and the capacity of the partner institutions to process and integrate them. Through this governance model, FIDA is positioned as a durable, evolving, and publicly oriented knowledge resource for Europe's film sector.

References

Zemaityte, V., Karjus, A., Rohn, U., Schich, M., & Ibrus, I. (2024). Quantifying the global film festival circuit: Networks, diversity, and public value creation. PLOS ONE, 19(3), e0297404. doi:10.1371/journal.pone.0297404

Annex 1

Data Processing per Data Source

The Movie Database (TMDB)

Data collection and Processing

The TMDB API v3 was employed to retrieve data, which consisted of JSON files with information about film productions including original title, genres, budget, TMDB ID, release date, revenue etc.

During preprocessing, this raw data was transformed and restructured into a table below to allow analysis. The columns updated_at and source were added to identify data source and information of when the data was last updated. Finally, a row hash was added to the rows to prevent double entries. One database was created:

- tmdb_raw_movie: containing all retrieved film metadata (original title, title, genres, budget, TMDB ID, release date, cast, crew, etc.)

This information would then be transformed in the Bronze Layer and become the backbone of the FIDA data pool.

Quality Control

Since TMDB, alongside Lumiere Pro, is one of the two key backbone components of FIDA, comprehensive Quality Control (QC) pipeline was followed to build structural and semantic integrity:

1. Ingestion and completeness control
    - Certain data-heavy information non-relevant to FIDA (e.g., video trailers, poster images, cast and crew images, etc.) were not selected for ingestion.
    - Blank or null-heavy records were pruned to save space, while making exceptions for film titles with explainable data gaps (such as titles not yet released).

2. Standardization of data
    - The meaning and usage of different ID codes was differentiated by re-naming some columns (e.g., cast_id, crew_id).
    - Verified the meaning of values in the field *gender* (0 = Not set / not specified. 1 = Female, 2 = Male, 3 = Non-binary)
    - Converted budgets recorded in USD ($) to EUR (€) using information from the World Bank.
    - Validated ISO country and language codes.
3. Join readiness
    - The availability of IDs against the backbone core database (Lumiere Pro) was tracked to support joins in the Silver and Gold Layers.

Bronze Layer (Raw Data Ingestion)

That data was ingested to the Bronze layer of the Databricks environment. The layer stored the data in Delta format, preserving its original characteristics to ensure traceability and updatability. One table was created:

Table 13: tmdb_raw_movie

Field Name	Data Type
adult	boolean
backdrop_path	string
belongs_to_collection	string
budget	bigint

genres: {"items": {"id": "int", "name": "string"}}	array
homepage	string
tmdb_id	int
imdb_id	string
origin_country: {"items": "string"}	array of strings
original_language	string
original_title	string
overview	string
popularity	float
poster_path	string
production_companies: {"items": {"id": "int", "logo_path": "string", "name": "string", "origin_country": "string"}}	array
production_countries: {"items": {"iso_3166_1": "string", "name": "string"}}	array
release_date	string
revenue	bigint
runtime	int

spoken_languages: {"items": {"english_name": "string", "iso_639_1": "string", "name": "string"}}	array
status	string
tagline	string
title	string
video	boolean
vote_average	float
vote_count	int
alternative_titles: {"titles": {"items": {"iso_3166_1": "string", "title": "string", "type": "string"}}}	struct
external_ids: {"imdb_id": "string", "wikidata_id": "string", "facebook_id": "string", "instagram_id": "string", "twitter_id": "string"}	struct
keywords: {"keywords": {"items": {"id": "int", "name": "string"}}}	struct
credits: {"cast": {"items": {"adult": "boolean", "cast_id": "int", "character": "string", "credit_id": "string", "gender": "int", "known_for_department": "string",	struct

"id": "int", "name": "string", "order": "int", "original_name": "string", "popularity": "float", "profile_path": "string"}}, "crew": {"items": {"adult": "boolean", "credit_id": "string", "department": "string", "gender": "int", "known_for_department": "string", "id": "int", "job": "string", "name": "string", "original_name": "string", "popularity": "float", "profile_path": "string"}}}	
release_dates: {"results": {"items": {"iso_3166_1": "string", "release_dates": {"items": {"certification": "string", "descriptors": {"items": "string"}, "iso_639_1": "string", "note": "string", "release_date": "string", "type": "int"}}}}}	struct
reviews: {"page": "int", "results": {"items": {"author": "string", "author_details": {"name": "string", "username": "string", "avatar_path": "string", "rating": "double"}, "content": "string", "created_at": "string", "id": "string", "updated_at": "string", "url": "string"}}, "total_pages": "int", "total_results": "int"}	struct
images: {"backdrops": {"items": {"aspect_ratio": "double", "height": "int", "iso_639_1": "string", "file_path": "string", "vote_average": "double", "vote_count": "int", "width": "int"}}, "logos": {"items": {"aspect_ratio": "double", "height": "int", "iso_639_1": "string", "file_path": "string", "vote_average": "double", "vote_count": "int", "width": "int"}}, "posters": {"items": {"aspect_ratio": "double", "height": "int", "iso_639_1": "string", "file_path": "string", "vote_average": "double", "vote_count": "int", "width": "int"}}}	struct
translations: {"translations": {"items": {"iso_3166_1": "string", "iso_639_1": "string", "name": "string", "english_name": "string", "data": {"homepage": "string", "overview": "string", "runtime": "int", "tagline": "string", "title": "string"}}}}	struct

videos: {"results": {"items": {"iso_639_1": "string", "iso_3166_1": "string", "name": "string", "key": "string", "site": "string", "size": "int", "type": "string", "official": "boolean", "published_at": "string", "id": "string"}}}	struct

The information was kept as it was ingested to ensure that it can be updated and changes can be backtracked.

Silver Layer (Transformation and Integration)

In the FIDA Silver Layer, the TMDB raw data was organized into a series of tables. The main table (tmdb_fact_movie) contains general film information, with the goal of it to become part of the FIDA backbone in the Gold Layer. Additional information that would for the other Gold Layer components was organized as separate tables resulting in the creation of ten tables with TMDB data.

Table 14: tmdb_fact_movie

This table contains basic film production information, and would be joined with LumierePro data to create the FIDA Master Table (*dim_filmmaster*) in the FIDA Gold Layer.

Name	Type	Description
tmdb_id	int	TMDB unique movie identifier
adult	boolean	Indicates if the movie is for adults
backdrop_path	string	Path to the backdrop image
belongs_to_collection	string	Collection the movie belongs to

budget	bigint	Movie budget amount
facebook_id	string	Facebook page ID or URL
homepage	string	Official homepage URL
imdb_id	string	IMDb identifier
instagram_id	string	Instagram page ID or handle
original_title	string	Original title of the movie
overview	string	Short description or synopsis
popularity	float	Popularity score from TMDB
poster_path	string	Path to the poster image
release_date	string	Movie release date
revenue	bigint	Total revenue earned
runtime	int	Movie duration in minutes
status	string	Current release status (e.g., Released, Post Production)
tagline	string	Movie tagline or slogan

Table 15: tmdb_alternative_titles

This table table contains alternative titles, and is not included in the Gold Layer. However it remains accessible via the TMDB ID.

Name	Type	Description
tmdb_id	int	TMDB unique identifier
iso_3166_1	string	Country code following ISO 3166-1 standard
title	string	Title or name associated with the record
type	string	Type or category of the entry
pos	int	Position or order value
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference
rowhash	string	Hash value representing the row's

		content

Table 16: tmdb_cast

This table table contains cast information, and is not included in the Gold Layer. However it remains accessible via the TMDB ID.

Name	Type	Description
tmdb_id	int	TMDB unique movie identifier
adult	boolean	Indicates if the person is an adult actor
cast_id	int	Unique cast ID within the movie credits
character	string	Character name played by the actor
credit_id	string	TMDB credit identifier
gender	string	Gender of the cast member
known_for_department	string	Department or field the person is known for (e.g., Acting, Directing)
id	int	TMDB person ID
name	string	Full name of the person

order	int	Order of appearance in credits
original_name	string	Original name of the person (if different from localized version)
popularity	float	Popularity score from TMDB
profile_path	string	Path to the profile image
pos	int	Position or order value
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference
rowhash	string	Hash value representing the row's content

Table 17: tmdb_crew

This table table contains crew information, and is not included in the Gold Layer. However it remains accessible via the TMDB ID.

Name	Type	Description
tmdb_id	int	TMDB unique movie identifier
adult	boolean	Indicates if the person is an adult contributor
credit_id	string	TMDB credit identifier
department	string	Department the person worked in (e.g., Directing, Writing, Editing)
gender	string	Gender of the crew member
known_for_department	string	Primary department or field the person is known for
id	int	TMDB person ID
job	string	Specific job title or role (e.g., Director, Producer)
name	string	Full name of the person
original_name	string	Original name of the person (if localized name differs)
popularity	float	Popularity score from TMDB

profile_path	string	Path to the profile image
pos	int	Position or order value
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference
rowhash	string	Hash value representing the row's content

Table 18: tmdb_genres

The information of this table was used to develop two Gold Layer tables:

- Main Genre (dim_main_genres), which list the genres that encompass all other listed genres of a film.
- Genres (dim_genres), which list all the available genres applicable to FIDA films.

Name	Type	Description
tmdb_id	int	TMDB unique identifier
id	int	Internal or related record

		identifier
name	string	Name or label associated with the record
pos	int	Position or order value
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference
rowhash	string	Hash value representing the row's content

Table 19: tmdb_production_companies

This table table contains production company information, and is not included in the Gold Layer. However it remains accessible via the TMDB ID.

Name	Type	Description

tmdb_id	int	TMDB unique movie identifier
id	int	Company or organization identifier
logo_path	string	Path to the company's logo image
name	string	Name of the company or production entity
origin_country	string	Country of origin (ISO 3166-1 code)
pos	int	Position or order value
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference
rowhash	string	Hash value representing the row's content

Table 20: tmdb_production_countries

This table table contains production country information. Some elements of this table were used to standardize the field of production country and were used to create the Gold Layer table for Production Country (*dim_productioncountries*).

Name	Type	Description
tmdb_id	int	TMDB unique movie identifier
iso_3166_1	string	Country code following ISO 3166-1 standard
name	string	Name of the country or region
pos	int	Position or order value
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference

rowhash	string	Hash value representing the row's content

Table 21: tmdb_release_dates

This table table contains release date information. As a film could have several release dates (e.g., world premiere, international premiere, domestic premiere, etc.) the information for this table was used to complete the related fields in the FIDA Master (*dim_filmmaster*) and Distributions tables (*dim_distributions*).

. Some elements of this table were used to standardize the field of production country and were used to create the Gold Layer table for Production Country (*dim_productioncountries*).

Name	Type	Description
tmdb_id	int	TMDB unique movie identifier
iso_3166_1	string	Country code following ISO 3166-1 standard
certification	string	Official content rating or certification (e.g., PG-13, R)
descriptors	array (items: string)	List of content descriptors or tags (e.g., "violence", "language")
iso_639_1	string	Language code following ISO 639-1 standard

note	string	Additional notes about the release or certification
release_date	string	Date of the movie's release
type	string	Type or category of the release (e.g., theatrical, digital)
meaning	string	Explanation or meaning of the certification or release type
order	bigint	Order or ranking of the record
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference
rowhash	string	Hash value representing the row's content

Table 22: tmdb_spoken_languages

This table table contains information about spoken languages, which was used to build the Gold Layer table Spoken Languages (*dim_spoken_languages*).

Name	Type	Description
Column Name	Data Type	Description
tmdb_id	int	TMDB unique movie identifier
english_name	string	English name of the language
iso_639_1	string	Language code following ISO 639-1 standard
name	string	Native name of the language
pos	int	Position or order value
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference

rowhash	string	Hash value representing the row's content

Table 23: tmdb_translations

This table table contains information regarding the translated titles, taglines and overview of a film production.

While this table is not included in the Gold Layer, it remains accessible in the Silver Layer via the TMDB ID.

Name	Type	Description
tmdb_id	int	TMDB unique movie identifier
iso_639_1	string	Language code following ISO 639-1 standard
iso_3166_1	string	Country code following ISO 3166-1 standard
name	string	Localized or alternative movie name
english_name	string	English version of the movie name
homepage	string	Official movie homepage URL
overview	string	Description or synopsis of the movie

runtime	int	Movie duration in minutes
tagline	string	Movie tagline or slogan
title	string	Official title of the movie
pos	int	Position or order value
unique_key	string	Unique key used for deduplication or joins
source	string	Data source or origin
loaded_at	timestamp	Timestamp when the data was initially loaded
updated_at	timestamp	Timestamp when the data was last updated
file	string	Source file name or reference
rowhash	string	Hash value representing the row's content

Gold Layer (Analytical Output)

The FIDA Gold Layer aggregates curated data into analytical models that support descriptive and predictive analyses. Thus, information that could identify single films was not ingested. The following TMDB data has been ingested into the Gold Layer data pool:

- Production data (e.g., year of production, budget, production countries)

- Temporal and location data (e.g., release dates by country)
- Cross-database enrichment (e.g., linking with other tables for admissions or box office metrics).

TMDB data in the film master table of the Gold layer helps to widen the data pool by including films not found within Lumiere Pro or Wikidata. Thus, it enriches the insights regarding film production output and lifecycle in smaller markets.

Wikidata

Data collection and Processing

The Wikidata API was used to create a data dump with film production information such as original title, genres, cast, crew, budget, release date, box office, etc. During preprocessing, the raw data was organised into tables.

The data retains the original column names and organisation of the source, in order to allow updates. Further transformations were performed to include certain data points as part of the FIDA backbone in the Gold Layer.

Quality Control

WikiData is a repository in which volunteer editors can add, remove, and update details. Therefore, the following activities were performed in order to verify the reliability of the data, remove identical duplicates, and merge multiple records about the same film:

1. Selection of trusted fields

- Certain fields were selected to prioritise them as sources of truth for specific film characteristics (e.g. international title instead of original title).

2. Data harmonisation
    - Some field formats (e.g. currency, dates, etc.) were standardize in order to facilitate film matchmaking and interoperability between data sources.
3. External validation
    - The IDs included in each record was cross-checked against the wider dataset created by joining Lumiere Pro and TMDB.
4. Redundancy elimination and deduplication
    - Identical records, or rows in a table with small variations of a particular field (e.g., film festival name) were consolidated into single entries.

Bronze Layer (Raw Data Ingestion)

After QC, the data was ingested into the Bronze layer of the Databricks environment. This layer stores the data in Delta format, preserving its semi-structured nature while ensuring traceability and reproducibility. Twenty-nine tables were created:

- Wiki_raw_sound_designer: containing the entry for the sound designer.
- Wiki_raw_actor: containing the entry for cast information.
- Wiki_raw_award_received: containing the entry for awards received.
- Wiki_raw_base: containing all basic film data (original title, title, genres, budget, Wiki ID, runtime, crew, etc.)
- Wiki_raw_boxoffice: containing the entry for box office.
- Wiki_raw_composer: containing the entry for film music composer.
- Wiki_raw_costumdesigner: containing the entry for costume designer.

- Wiki_raw_director: containing the entry for film director.
- Wiki_raw_director_photography: containing the entry for director of photography.
- Wiki_raw_distributed: containing the entry for film distributor.
- Wiki_raw_executive_producer: containing the entry for executive producer.
- Wiki_raw_filmeditor: containing the entry for film editor.
- Wiki_raw_filming_location: containing the entry for filming location.
- Wiki_raw_follows: containing the entry for preceding or related work.
- Wiki_raw_genre: containing the entry for film genre.
- Wiki_raw_identifier: containing the entry for unique identifier.
- Wiki_raw_main_subject: containing the entry for main subject.
- Wiki_raw_movie_language: containing the entry for movie language.
- Wiki_raw_narrative_location: containing the entry for narrative location.
- Wiki_raw_nominated_for: containing the entry for award nomination.
- Wiki_raw_original_language: containing the entry for original language.
- Wiki_raw_part_of_series: containing the entry for film series membership.
- Wiki_raw_presented_in: containing the entry for presentation format or medium.
- Wiki_raw_producer: containing the entry for film producer.
- Wiki_raw_production_company: containing the entry for production company.
- Wiki_raw_production_designer: containing the entry for production designer.
- Wiki_raw_publication: containing the entry for publication or release.
- Wiki_raw_reviews: containing the entry for film reviews.
- Wiki_raw_screenwriter: containing the entry for screenwriter.

Silver Layer (Transformation and Integration)

Data was transformed within the Silver layer in order to create tables that supported analytical queries. Certain fields names were harmonised against other fields belonging to external sources in order to ease matchmaking . When ID matchmaking was not possible, fuzzy string matching was performed.

WikiData tables were joined in order to create an additional table containing basic film production information through ID numbers which connect this master table to other dimension tables.

Table 24: wiki_fact_movies

Column Name	Data Type	Description
id	string	Unique identifier for the film.
id_actor	string	Identifier of actor(s) in the film.
id_director	string	Identifier of the film's director.
id_distributed	string	Identifier of the distribution company or distributor.
duration	string	Runtime or total length of the film
id_genre	string	Identifier for the film's genre
id_main_subject	string	Identifier for the main subject or theme of the film.
id_original_language	string	Identifier for the original language of the film.
id_director_photography	string	Identifier for the director of photography

id_executive_producer	string	Identifier for the executive producer(s).
id_filmeditor	string	Identifier for the film editor.
id_filming_location	string	Identifier for where the film was shot.
id_narrative_location	string	Identifier for the story's setting or narrative location.
id_producer	string	Identifier for the film's producer(s).
id_screenwriter	string	Identifier for the screenwriter(s).
id_composer	string	Identifier for the composer of the film's score.
id_costumdesigner	string	Identifier for the costume designer.
id_production_designer	string	Identifier for the production designer.
costs	int	Production budget.
id_part_of_series	string	Identifier for the series the film is part of, if any.
id_production_company	string	Identifier for the production company.
id_follows	string	Identifier of the preceding film, if any.
start_time	date	Date when filming started.
end_time	date	Date when filming ended.
premiere	date	Date of the film's first public showing or release.

source	string	Source of the data.
loaded_at	timestamp	Timestamp when the data was first loaded.
updated_at	timestamp	Timestamp when the data was last updated.
rowhash	string	Hash value used for deduplication or data integrity.

Gold Layer (Analytical Output)

The Gold Layer joins curated WikiData data into the Master Table and dimension tables of FIDA. This includes:

- Market-level KPIs (e.g., year of release, countries of distribution per title);
- Production information (e.g. budget, cast, crew, etc.)
- Cross-database enrichment (e.g., Film IDs for a variety of platforms).

By using WikiData as a secondary source to complement the data backbone composed of TMDB and Lumiere Pro information, FIDA extends its data pool to include details that provide context to the observations made across film markets.

World Bank

Data collection and Processing

The World Bank data consisted of a single Excel file:

- Worldbank-population: containing the full name, ISO code, and population of every country listed in the World Bank database in the years 1990, 2000, and 2014 to 2023.

During the pre-processing stage the table and field names were kept as in the raw data, in order to keep consistency in future updates.

Quality Control

The data from the World Bank was already organized, yet the following steps were taken to ensure reliability:

1. Data harmonisation

- Countries with multiple records (due to country name changes) were identified. This information would be used to build a unified country list in the Silver Layer.

2. Consistency check

- The data across years within a single country was compared to detect anomalies or missing values.

3. Validation of format

- The data was reviewed in order to confirm it was properly organized, with standardized country names, ISO country codes, and consistent number formats.

Through this process, it was possible to validate that the data was complete, organized, and ready for further processing.

Bronze Layer (Raw Data Ingestion)

The World Bank data was ingested into the Bronze layer of the Databricks environment in Delta format, ensuring traceability and reproducibility.

Table 25: crescine.bronze.worldbank-population

Column	Type
Series Name	string
Series Code	string
Country Name	string

Country Code	string
1990 [YR1990]	bigint
2000 [YR2000]	bigint
2014 [YR2014]	bigint
2015 [YR2015]	bigint
2016 [YR2016]	bigint
2017 [YR2017]	bigint
2018 [YR2018]	bigint
2019 [YR2019]	bigint
2020 [YR2020]	bigint
2021 [YR2021]	bigint
2022 [YR2022]	bigint
2023 [YR2023]	bigint

At this stage, data was retained in near-raw form to preserve original fidelity and enable backtracking if needed.

Silver Layer (Transformation and Integration)

In this stage, column names were reformatted to enhance readability while country names were standardized by adding the column *country_as_used_in_crescine*. This created a redundant way to connect to the table by either using this field or the country code. In addition, timestamps were added in order to log data updates and detail version history.

Table 26: crescine.silver.worldbank_population

Column	Type
Country_name	string
country_as_used_in_crescine	string
Country_code	string
Year_1990	bigint
Year_2000	double
Year_2014	double
Year_2015	double
Year_2016	double
Year_2017	double
Year_2018	double

Year_2019	bigint
Year_2020	double
Year_2021	bigint
Year_2022	double
Year_2023	bigint
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Gold Layer (Analytical Output)

The Silver Layer data was ported into the Gold Layer, and connects to the main via the standardized column *country_as_used_in_crescine*. The World Bank information in Gold Layer allows for the following types of queries:

- Time and Location (e.g., year and country)
- Population
- Cross-database enrichment (e.g., linking with other tables for per capita and country-level calculations).

The inclusion of this World Bank data enables the contextualization of analyses that examine the performance of films within a particular country in a specific year, as well as enabling the per-capita calculations needed to compare the economic performance of different film markets.

Cinando

Data collection and Processing

The collection and preprocessing workflow transformed a MS Access data dump provided by Cinando into structured, analyzable data. During the pre-processing stage, labeling conventions, table names, and file names from the raw data were kept to create consistency for future data updates.

The raw data was split across tables, which could be linked and merged using the cFilm field for films, and ref* variables for various film properties (e.g. country, company) with more data in their respective tables.

The data convention in the provided Cinando dump is unusual. Fields have inconsistent names between larger and smaller metadata tables. The variable isCinando variable indicates whether the data point was entered and verified by Cinando employees (1) or platform users (0, the majority). Smaller Cinando tables have a NoOrder variable, which roughly indicates priority (e.g. primary production country) but there is no guarantee users have used this properly. Film IDs were duplicated if they had multiple categories. Finally, there is a variable (Active or Actif) which identifies (with a 0) if an entry was deleted. Thus it is recommended that only Actif=1 entries are used.

Below are short summaries for the tables exported from the MS Access database dump provided by Cinando

Seventeen tables were created within a single volume:

- Countries: contains countries plus ID codes for linking.
- Event (temp): is the general festival events table. It contains information such as event name, event ID code, series ID, location country, and dates. [2]
- Events (main): contains the festival series present in the database.
- Country (film): contains films and their main production country.
- Festivals (temp): this is a merged table of films at festivals.
- Film Crew: contains the names crew members and roles per film, with the roles of producer and director being the most likely ones present.
- Films (Films Roles): contains information about the film production companies.
- Films (generic): this is another film crew table, seems to contain the same data as the above, but without names and few other variables.
- Kind: contains films and their genres, themes, and other variable tags referring to the director (e.g. first film) or technical aspects (e.g. VR, 3D).
- Market history: contains details about a film's participation at film markets.
- Match film: contains meta-information of when data was added and by whom.
- Media: contains information about media assets (e.g. posters) associated with films.
- Films (main): contains cFilm ID that can be used to link to other tables, original film title, alternative title, completion status, and year of production.
- Films (generic): contains the same data as the above, but lacks names and few other variables.

[2] Note: As the data from Cinando is user-submitted, there is a considerable number of duplicate events. For example, Cannes Film Festival appears under several slightly different names and different event IDs. This table can be linked using the festival series(idFestival) and libelleFestival (title) columns. While most events are not attached to a festival series, this does not rule out said possibility.

- TIDX Film: contains the link to identify which film appeared at which festival, awards received, name of the award, and place where an award was given.
- TIDX Film Language: contains film languages
- Film language: contains film languages and other fields about languages such as language codes.

These tables were later ingested in the Bronze layer into the Databricks data architecture.

Quality Control

Given the heterogeneity of the raw API responses, a comprehensive Quality Control (QC) pipeline was implemented to ensure structural and semantic integrity.

1. Structural validation
    - Duplicated records were filtered and merged.
2. Data hierarchisation
    - Meaningful labels (e.g. full country names) were contained in smaller tables, with the main Cinando tables referring to them via c/ref codes.
3. External validation
    - The field *eidr_id* was cross-checked against the eidr from the wider dataset created by joining Lumiere Pro and TMDB.
4. Redundancy elimination and deduplication
    - Identical records, or rows in a table with repeated variations of a particular field (e.g., film festival name) were consolidated into a single entry.

Through these measures, the dataset achieved both internal coherence and external referential integrity.

Bronze layer (Raw Data Ingestion)

After QC, the data was ingested into the Bronze layer of the Databricks environment. This layer stores the data in Delta format, preserving its semi-structured nature while ensuring traceability and reproducibility.

Seventeen tables were created:

Table 27: crescine.bronze.cinando_raw_tcountries

Column	Type
cCountry	bigint
NameEN	string
cContinent	bigint
Continent	string
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 28: crescine.bronze.cinando_raw_tevent_temp_20250115

Column	Type
UID	bigint
VID	double
CREEPAR	bigint

MODIFIEPAR	double
DATECREATION	string
DATEMODIFICATION	double
DateModificationAdm	double
refUtilisateurAdm	double
ACTIF	bigint
cEvent	bigint
refTypeEvent	bigint
idFestival	double
LibelleEvent	string
DateDebut	string
DateFin	string
Year	double
refMarketEdition	double
isCinando	bigint
isFocusCatchup	double
IstTypeEvent	double

Table 29: cinando_raw_teventmain_temp_20250115

Column	Type
idFestival	bigint
libelleFestival	string
refCompany	double

rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 30: crescine.bronze.cinando_raw_tfilms_country_temp

Column	Type
cFilm	bigint
refCountry	bigint
Main_Country	string
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 31: crescine.bronze.cinando_raw_tfilms_festivals_temp

Column	Type

cFilm	bigint
TitleVA	string
refCompany	bigint
Company	string
idFestival	double
libelleFestival	string
refEvent	bigint
Year	double
isCinando	bigint
IstTypeEvent	double
refEventSection	double
LibelleSection	string
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp

file	string

Table 32: crescine.bronze.cinando_raw_tfilms_filmcrew_temp

Column	Type
cFilm	bigint
refFilmCrew	bigint
IstType	bigint
txtType	string
NoOrder	double
FirstName	string
LastName	string
Company	string
Actif	bigint
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp

file	string

Table 33: crescine.bronze.cinando_raw_tfilms_filmsroles_temp

Column	Type
cFilm	bigint
TitleVA	string
fVis	bigint
IstProductionStatus	double
txtProductionStatus	string
refCompany	bigint
Company	string
ceVis	bigint
SALES	bigint
DISTR	bigint
PROD	bigint
FIN	bigint
FEST	bigint

PROMO	bigint
REP	bigint
CONS	bigint
BROAD	bigint
rowhash	string
source	string
loaded_at	timestamp

Table 34: cinando_raw_tfilms_generic_temp

Column	Type
cFilm	bigint
refFilmCrew	bigint
IstType	bigint
txtType	string
NoOrder	double
Actif	bigint
rowhash	string

source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 35: crescine.bronze.cinando_raw_tfilms_kind_temp

Column	Type
Active	comp
cFilm	bigint
Main_Genre	string
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 36: crescine.bronze.cinando_raw_tfilms_markethistory_temp

Column	Type

cFilm	bigint
TitleVA	string
refCompany	bigint
Company	string
refMarketEdition	bigint
Year	bigint
cMarket	bigint
Name	string
IstType	double
txtType	string
IstMarketStatus	double
txtMarketStatus	string
historicStatus	double
IstProductionStatus	double
txtProductionStatus	string
rowhash	string

source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 37: crescine.bronze.cinando_raw_tfilms_match_film_temp

Column	Type
IDFILM	bigint
REFFILMPARTENAIRE	bigint
IDPARTENAIRE	bigint
ISPROJET	double
ANNEEPRODUCTION	double
ISAN	string
ACTIF	bigint
DATECREATION	string
DATEMODIFICATION	string
CREEPAR	string

MODIFIEPAR	string
REFMATCHING	double
LSTTYPETRAITEMENT	bigint
TXTTYPETRAITEMENT	string
COMMENTAIREELEMENT	string
ISHORSTRAITEMENT	bigint
ISENCOURSTRAITEMENT	bigint
TRAITEPAR	string
DATEDEBUTTRAITEMENT	double
rowhash	string

Table 38: crescine.bronze.cinando_raw_tfilms_media_temp

Column	Type
cFilm	bigint
dbo_idxFilmMedia_Actif	bigint
refMedia	bigint
dbo_Media_Actif	bigint

IstType	bigint
txtType	string
Status	double
ProcessId	double
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 39: crescine.bronze.cinando_raw_tfilms_temp

Column	Type
cFilm	bigint
TitleVA	string
fVis	bigint
IstTypeFilm	bigint
txtTypeFilm	string

IstCategory	double
txtCategory	string
IstProductionStatus	bigint
txtProductionStatus	string
refCountry	bigint
refCompany	bigint
Company	string
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 40: crescine.bronze.cinando_raw_tfilms_temp_20250115

Column	Type
value	string

Table 41: crescine.bronze.cinando_raw_tidxfilmevent_temp_20250115

Column	Type
uid	bigint
DateCreation	string
CreePar	bigint
refEvent	bigint
refFilm	bigint
refEventSection	double
awardDescription	string
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 42: crescine.bronze.cinando_raw_tidxfilmlanguage_temp_20250115

Column	Type
uid	bigint

vid	double
CreePar	bigint
ModifiePar	double
DateCreation	string
DateModification	string
DateModificationAdm	string
refUtilisateurAdm	double
Actif	bigint
EtatFiche	double
refFilm	bigint
refLanguage	bigint
txtLanguage	string
NoOrder	bigint
rowhash	string
source	string
loaded_at	timestamp

updated_at	timestamp
file	string

Table 43: crescine.bronze.cinando_raw_tidxgeneric_temp_20250115

Column	Type
uid	bigint
vid	double
CreePar	bigint
ModifiePar	double
DateCreation	string
DateModification	string
DateModificationAdm	string
refUtilisateurAdm	double
Actif	bigint
EtatFiche	double
refFilm	bigint
refFilmCrew	bigint

IstType	bigint
txtType	string
NoOrder	double
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Table 44: crescine.bronze.cinando_raw_tidxgeneric_temp_20250115

Column	Type
uid	bigint
vid	double
CreePar	bigint
ModifiePar	double
DateCreation	string
DateModification	string

DateModificationAdm	string
refUtilisateurAdm	double
Actif	bigint
EtatFiche	double
cLanguage	bigint
refCountry	double
NameFR	string
NameEN	string
Code	string
isActifListingFilms	bigint
isActifListingFilms2	bigint
Name_1	string
Name_2	string
Name_3	double

At this stage, data were minimally processed and retained in near-raw form to preserve original fidelity and enable backtracking if needed.

Silver Layer (Transformation and Integration)

The Silver Layer consolidated the raw data into tables suitable for analytical queries and integration with external systems. To create the tables, data fields between Cinando and the master tables were normalized on both sides before matching. Three key transformation outputs were obtained:

- cinando_clean_hasduplicates: Contains variables relevant for matching with other databases. It only includes entries classified as "Actif", completed productions, or that have a production year. To ease linking the table contains ISAN, EIDR and the field refFilm (standardized as cinando_id). This table intentionally has duplicates as in the original database, which should be removed in matching, keeping the more complete entries. Different entries may link to different festivals though. The table contains a variable number_of_festivals that indicates how many entries are linked to a given refFilm ID in the festivals table. Please note that the same film can appear with multiple IDs in this table.
- cinando_films_filmmaster_joined: This table is a combination of the film master data and Cinando film data. The table contains different columns that enable it to be linked to other tables: cinando_id column allows for joining with other Cinando data objects (called refFilm as in the original source), crescine_id allows linking to all other Crescine datasets, and lumiere_id enables links to Lumiere and CresCine awards data, since it also contains it.
- cinando_films_festivals_filmmaster_joined: this table contains data on Cinando festivals that can be linked back to any other Crescine data via the crescine_id or to CresCine's award data via the lumiere_id, for further analysis.

Table 45: cinando_clean_hasduplicates

Column	Type	Description
title_original	string	

title_english	string	
production_year	bigint	
duration	string	
cinando_id	bigint	
refFilm	bigint	
Isan	string	
Eidr	string	
txtTypeFilm	string	
txtCategory	string	
logline	string	
synopsis	string	
spoken_language_main	string	
spoken_languages	string	
genres	string	
production_country_main	string	

Table 46: cinando_films_filmmaster_joined

Column	Type	Description
cinando_id	int	
crescine_id	int	
title_original_cinando	string	
lumiere_id	string	
title_original	string	
title_english	string	
production_year	int	
production_country_main	string	

Table 3: cinando_films_festivals_filmmaster_joined

Column	Type	Description
crescine_id	bigint	
lumiere_id	string	
cinando_id	bigint	
refEvent	bigint	
refEventSection	string	

awardDescription	string	
LibelleEvent	string	
idFestival	string	
txtCountry	string	
Year	string	
title_original_cinando	string	
title_original	string	
title_english	string	
production_year	bigint	
production_country_main	string	

Gold Layer (Analytical Output)

The Gold layer aggregates curated data into analytical models that support descriptive and predictive analytics within FIDA. The Cinando information from the Silver Layer is integrated within the film master table.

The Gold layer integrates the following CInando data:

- Temporal and location data (e.g., event years, festival countries)
- Industry-level KPIs (e.g., festival selections, awards per title);
- Cross-database enrichment (e.g., linking with other tables for admissions or screening metrics).

The inclusion of Cinando data in the film master table of the Gold layer enables the creation of analytical dashboardsand modeling pipelines that examine festival participation, awards performance, and festival screening cycles across markets.

CresCine Festival Data

Data collection and Processing

In FIDA, the processing of CresCine Festival Data refers to the transformation of three Excel files into analyzable data. The Excel file was based on data collected for the period 2014-2023 via web scrapper Octoparse from the festival web pages, festival web transcriptions if scrapping was not possible, and Wikipedia if the information was not available in the former two. The original Excel had two tabs:

- Participations and Nominations, containing festival, edition, section, title, director, production year.
- Awards, containing festival, edition, award, title, director, production year.

Quality Control

The QC stage for CresCine Festival aimed to maximize matchmaking between this source and the film master table, by performing:

1. Combining tables
    - The two tabs from the original file were combined in a master film list, with columns covering each of the fields previously separated in different tabs.
2. Matchmaking of incomplete records.
    - Matchmaking was performed via the Lumiere ID or via fuzzy matching.
    - To ensure that identifiers were consistent, Lumiere IDs in the original file were cross-checked with LumierePro identifiers.

3. Redundancy elimination and deduplication:
    - Identical records with the same fields(international_title, directors, country, platform, available_from, and available_to) were reduced to a single instance.

Bronze Layer (Raw Data Ingestion)

Data was ingested into the Bronze layer of the Databricks environment in a separate volume. This layer stores the data in Delta format, preserving its semi-structured nature.

Table 47: raw_awards

Field	Type	Description
Festival	string	Name of the festival where the film was shown.
SFI Scoring	string	Scoring or rating given by SFI (if applicable).
Edition	bigint	Edition number of the festival.
Section	string	Festival section or program the film belongs to.
Award	string	Awards the film has received (if any).
English Title	string	Official English title of the film.
Original Title	string	Title of the film in its original language.
Director	string	Name of the director.
Jury Status	string	Film's status regarding jury selection or competition.

Lumiere ID	string	Lumiere database identifier for the film.
Ecosystem	string	Ecosystem classification (industry, market, etc.).
Prod. Year	bigint	Year the film was produced.
Prod. Country	string	Country or countries where the film was produced.
Director(s)	string	Full list of directors.
Other Titles	string	Alternative or working titles.
Description	string	A summary or synopsis of the film.

Silver Layer (Transformation and Integration)

In the Silver Layer, the CresCine Festival table was enriched with a row hash for identification and trace back changes.

Gold Layer (Analytical Output)

The Gold layer aggregates curated data into analytical models that support descriptive and predictive analytics within FIDA. The CresCine information from the Silver Layer is integrated within the film master table.

The Gold layer integrates the following data:

- Temporal and location data (e.g., event years, festival name, edition)

- Industry-level KPIs (e.g., festival selections, awards per title);

The inclusion of CresCine festival data in the film master table of the Gold layer enables the creation of analytical dashboards and modeling pipelines that examine festival participation and awards performance across markets.

UsherU

Data collection and Processing

The collection and preprocessing workflow transformed the raw data into analyzable data. The data was received as a CSV export form UsherU, and contained information regarding the films, platforms, and catalogue availability.

Quality Control

The QC stage for UsherU aimed to maximize matchmaking between this source and the film master table, by performing:

4. Structural validation
    - Records with malformed or missing fields were removed.
5. Matchmaking of incomplete records.
    - Fuzzy string matching was performed for films missing an ID. By matching records via film characteristics (e.g. international title x year x country of origin triplet) it was possible to match incomplete records against the wider film master table.
6. External validation
    - To ensure that identifiers were consistent, retrieved tmdb_id and imdb_id were cross-checked through external APIs (notably TMDB).

7. Redundancy elimination and deduplication:
    - Identical records with the same fields(international_title, directors, country, platform, available_from, and available_to) were reduced to a single instance.

Through these measures, the dataset achieved both internal coherence and external referential integrity.

Bronze Layer (Raw Data Ingestion)

Data was ingested into the Bronze layer of the Databricks environment in a separate volume. This layer stores the data in CSV format, preserving its semi-structured nature while ensuring traceability and reproducibility.

Table 48: usheru

Column	Type	Description
international_title	string	International film title in English
original_title	string	Film title in original language
imdb	string	IMDB identifier
eidr	string	EIDR identifier
directors	string	Name of director
release_date	timestamp	Platform release date
platform	string	Name of Streaming Platform

country	string	Country of distribution
url	string	Link to catalogue
available_from	timestamp	Date of release in a platform
available_to	timestamp	Date of release in a platform
delivery_type	string	Type of streaming service

Silver Layer (Transformation and Integration)

The Silver Layer consolidates and standardizes the raw data into relational tables suitable for analytical queries and integration with external systems.

Table 49: usheru_fact_movie

Contains harmonized cinema-level metadata for geographical and organizational analysis.

Column	Type	Description
international_title	string	International film title in English
original_title	string	Film title in original language
imdb	string	IMDB identifier
eidr	string	EIDR identifier
directors	string	Name of director
release_date	timestamp	Platform release date

platform	string	Name of Streaming Platform
country	string	Country of distribution
url	string	Link to catalogue
available_from	timestamp	Date of release in a platform
available_to	timestamp	Date of release in a platform
delivery_type	string	Type of streaming service
unique_key	string	
rowhash	string	
source	string	
loaded_at	timestamp	
updated_at	timestamp	
file	string	

Gold Layer (Analytical Output)

The Gold layer aggregates curated data into analytical models that support descriptive and predictive analytics within FIDA. The Gold layer integrates:

- Market-level KPIs (e.g., platforms per title, countries of distribution per title);
- Cross-database enrichment (e.g., linking with TMDB for genre and popularity metrics).

This layer powers analytical dashboards, and modeling pipelines that examine streaming distribution, platform density, and temporal exhibition trends across markets.

LumierePro

Data collection and Processing

Market information data was exported from the Lumiere Pro web service and then ingested into Databricks. The labeling convention, table names, and file names from the raw data was maintained, in order to create consistency for future data updates.

Data was organised into six tables covering different aspects of market performance:

- lumiere_pro_country_iso_codes: contains country ISO codes.
- lumierepro_market_data: contains location data
- lumierepro_ranking: contains box office data
- lumierepro_raw_admissions: contains admissions data
- lumierepro_raw_box_office:contains box office data
- lumierepro_raw_movies: contains basic film information details (e.g., title, year, genres, etc.).

This was the foundation of the Bronze Layer in the Databricks architecture.

Quality Control

To ensure interoperability, the following Quality Control actions were performed:

1. Elimination of incomplete records
    - Malformed records and rows with null values were eliminated.
2. Data harmonisation

- Some field formats (e.g. currency, dates, order of production country etc.) were standardized to match other data sources.
- Schemas were audited to identify and correct naming convention inconsistencies

3. Data organisation.
    - Columns featuring information for different fields were split into separate columns.
4. External validation
    - The IDs included in each record were cross-checked against TMDB in order to ensure coherence.

With these measures, the ingested data was ready for future transformations.

Bronze Layer (Raw Data Ingestion)

Data ingested into the Bronze layer of the Databricks environment was stored in Delta format. The ingested tables preserve the semi-structured nature of the raw data while ensuring traceability and reproducibility.

Table 50: lumiere_pro_country_iso_codes

Field Name	Data Type	Description
ISO code	string	ISO country code
Country	string	Country name
All countries	string	Identifies country as part of the group “All countries”
Asia	string	Identifies a country as part of the group of countries located in Asia.

European Union after Brexit	string	Identifies a country as part of the group of countries located in Europe after Brexit.
European Union before Brexit	string	Identifies a country as part of the group of countries located in Europe before Brexit.
Europe (CoE)	string	Identifies a country as part of the group of countries located in Europe (CoE).
Latin America & Caribbean	string	Identifies a country as part of the group of countries located in Latin America & Caribbean.
Middle East & North Africa (MENA)	string	Identifies a country as part of the group of countries located in Middle East & North Africa (MENA).
North America	string	Identifies a country as part of the group of countries located in North America.
Oceania	string	Identifies a country as part of the group of countries located in Oceania.
Other Europe (non-CoE)	string	Identifies a country as part of the group of countries located in Other Europe (non-CoE).
Sub-Saharan Africa	string	Identifies a country as part of the group of

		countries located in Sub-Saharan Africa.
United Kingdom + Ireland	string	Identifies a country as part of the group of countries located in the United Kingdom + Ireland.

Table 51: lumierepro_market_data

Field Name	Data Type	Description
year	bigint	Year
market	string	Market country.
region	string	Market country region.
providers	string	Data provider (e.g., BFI, MEDIA, Screen Ireland, etc.)
nb_films	bigint	Amount of films
admissions	bigint	Admissions.
coverage	double	
timestamp	timestamp	Data in which the data was loaded.

Table 52: lumierepro_ranking

Field Name	Data Type	Description

rank	int	
lumiere_id	int	Lumiere ID
imdb_id	string	IMDB ID
tmdb_id	string	TMDB ID
market	string	Market country
year	int	
original_title	string	Film title (original)
prod_year	int	Production year
prod_country	string	Production country
directors_gender	string	
box_office_eur	int	Box office in euros.

Table 53: lumierepro_raw_admissions

Field Name	Data Type	Description
admissions	bigint	Admissions
year	bigint	Year
market	string	Market country

national	boolean	
region	string	
id	bigint	
timestamp	timestamp	

Table 54: lumierepro_raw_box_office

Field Name	Data Type	Description
rank	int	
lumiere_id	int	Lumiere ID
imdb_id	string	IMDB ID
tmdb_id	string	TMDB ID
market	string	Market country
original_title	string	
prod_year	int	Production year
prod_country	string	Production country
directors_gender	string	Film director's genre
box_office_eur	int	Box Office in euros (€)

Table 55: lumierepro_raw_movies

Field Name	Data Type	Description
id	int	
original_title	string	
production_countries	string	
directors	string	
other_titles	string	
links	string	
distributions	string	
total_admissions_obs	int	
genres	string	
prod_year	int	

At this stage, data were minimally processed and retained in near-raw form to preserve original fidelity and enable backtracking if needed.

Silver Layer (Transformation and Integration)

The Silver Layer consolidates and standardizes Lumiere Pro information in tables suitable for analytical queries and integration with external systems. Each record was enriched with a row hash for identification and trace back changes. Eight different tables were created:

Table 56: lumierepro_admissions

Field Name	Data Type	Description
id	int	
year	int	
market	string	Market country
national	boolean	
national_extended	boolean	
region	string	Market country region.
admissions	int	Total of Admissions
created_at	timestamp	Date of creation.
rowhash	string	

Table 57: lumierepro_box_office

Field Name	Data Type	Description
lumiere_id	int	
imdb_id	string	
tmdb_id	string	

market	string	
year	int	
prod_year	int	
prod_country	string	
directors_gender	string	
box_office_eur	int	

Table 58: lumierepro_countries_iso

In the Silver Layer, this table change the data type used to mark a country as part of group from string (X) to boolean (True, False) values.

Field Name	Data Type	Description
iso_code	string	ISO country code
country	string	Country name
all_countries	boolean	Identifies country as part of the group "All countries"
asia	boolean	Identifies a country as part of the group of countries located in Asia.
eu_after_brexit	boolean	Identifies a country as part of the group of countries located in Europe after Brexit.

eu_before_brexit	boolean	Identifies a country as part of the group of countries located in Europe before Brexit.
europe_coe	boolean	Identifies a country as part of the group of countries located in Europe (CoE).
latin_america_caribbean	boolean	Identifies a country as part of the group of countries located in Latin America & Caribbean.
middle_east_north_africa_mena	boolean	Identifies a country as part of the group of countries located in Middle East & North Africa (MENA).
north_america	boolean	Identifies a country as part of the group of countries located in North America.
oceania	boolean	Identifies a country as part of the group of countries located in Oceania.
europe_other_non_coe	boolean	Identifies a country as part of the group of countries located in Other Europe (non-CoE).
sub_saharan_africa	boolean	Identifies a country as part of the group of countries located in Sub-Saharan Africa.
uk_ireland	boolean	Identifies a country as part of the group of countries located in the United Kingdom +

		Ireland.

Table 59: lumierepro_distributions

Field Name	Data Type	Description
id	int	
release_date	date	
first_release_date_movie	date	
last_release_date_movie	date	
company	string	
country	string	
rowhash	string	

Table 60: lumierepro_market_data

Field Name	Data Type	Description
market	string	
year	int	
region	string	
providers	string	

nb_films	int	
admissions	bigint	
coverage	double	
created_at	timestamp	
rowhash	string	

Table 61: lumierepro_movies

Field Name	Data Type	Description
id	int	
imdb_id	string	
original_title	string	
directors	string	
other_titles	string	
url_and_external_id	string	
distributions	string	
total_admissions_obs	int	
film_type	string	

film_make	string	
obs_genre	string	
prod_year	int	
directors_gender	string	
rowhash	string	
created_at	timestamp	
wiki_id	string	

Table 62: lumierepro_other_titles

Field Name	Data Type	Description
id	int	
name	string	
is_original	boolean	
country_code	string	
created_at	timestamp	
rowhash	string	

Table 63: lumierepro_production_countries

Field Name	Data Type	Description
id	int	
production_country	string	
order_of_appearance	int	
number_of_production_countries	int	
created_at	timestamp	
rowhash	string	

In addition to these tables which included only Lumiere Pro data, a joint table was created in order to create a first version of the Master Table of the FIDA Gold Layer.

Table 64: movies_similarity_joined

Field Name	Data Type	Description
title_original	string	
title_english	string	
production_year	double	
production_country_main	string	
release_date	string	

lumiere_id	double	
tmdb_id	int	
wiki_id	string	
imdb_id	string	

Gold Layer (Analytical Output)

The Gold layer aggregates curated data into analytical models that support descriptive and predictive analytics within FIDA. Lumiere Pro is one of the two backbone components of the FIDA Gold Layer by providing:

- Market-level KPIs (e.g., admissions, box-office);
- Location information (e.g., admissions country, country of origin):
- Cross-database interconnectivity (e.g., TMDB ID, Lumiere ID, IMDB ID).

Thus, Lumiere Pro is one of the two critical components that delimit the FIDA data pool as well as a critical element which enables interconnecting other databases ingested.

International Showtimes

Data collection and Processing

The collection and preprocessing workflow involved multiple stages to transform raw API responses into structured, analyzable data. Initially, data were retrieved from the Showtimes International API, which returns semi-structured JSON documents containing information about movies, cinemas, and showtimes.

The Movie Matcher service, also provided by Showtimes International, was used to retrieve unique movie_id values by querying the endpoint with the movie title. The results typically include additional metadata such as alternative titles, directors, release dates, and, when available, the tmdb_id.

Raw responses were stored in local JSON files and subsequently imported into a MongoDB database via a custom local API, enabling controlled ingestion into the Databricks data lake. During this phase, redundancy elimination was applied to remove multiple identical entries for the same (cinema × movie × screening) triplet.

Two base MongoDB collections were created:

- cinema – containing all known cinema metadata (name, address, country, coordinates, etc.);
- showings – containing detailed screening information with movie references and timestamps.

These formed the foundation for the Bronze layer in the Databricks data architecture.

Quality Control

Given the heterogeneity of the raw API responses, a comprehensive Quality Control (QC) pipeline was implemented to ensure structural and semantic integrity.

1. Structural Validation:

   Records with malformed or missing fields–particularly inconsistent or invalid start_at timestamps– were filtered and stored in a dedicated invalid__ directory.

2. Identifier Enrichment:

Since raw data often lacked identifiers (movie_id, cinema_id, and showtimes_id), the QC process integrated API calls to retrieve missing information:

- The Movie Matcher endpoint was queried repeatedly until a valid movie_id was obtained.
- For cinema identification, a fuzzy matching pipeline was applied, comparing attributes such as latitude, longitude, slug, address, and zipcode against the official Showtimes cinema registry per country.
    - Geolocation-based matching was prioritized when valid coordinates were available.
    - Fuzzy string matching was used when only textual data existed, with similarity thresholds of 100% for slug and zipcode, and 95% for address.

3. External Validation:

   Retrieved tmdb_id and imdb_id were cross-checked through external APIs (notably TMDB) to validate identifier consistency.

4. Redundancy Elimination and Deduplication:

   Identical screenings—defined by the same (country, movie_id, cinema_id, start_at)—were reduced to a single instance (showings).

Through these measures, the dataset achieved both internal coherence and external referential integrity.

Bronze Layer (Raw Data Ingestion)

Data ingested into the Bronze layer of the Databricks environment was stored in Delta format. The ingested tables preserve the semi-structured nature of the raw data while ensuring traceability and reproducibility.

Two main tables were created:

Table 65: showtimes_raw_cinemas

Column	Type	Description
cinema_id	string	Unique cinema identifier
name	string	Cinema name
address	string	Cinema address
country	string	Country code
lat	double	Latitude
lon	double	Longitude
slug	string	URL-friendly cinema identifier
chain	struct	Nested JSON with chain_id
loaded_at	timestamp	Data ingestion timestamp

Table 66: showtimes_raw_showings

Column	Type	Description
show_id	string	Unique showing identifier
date	string	Screening date
cinema	struct	Contains cinema_id and country
movie	struct	Contains is_id, title, tmdb_id,

		imdb_id
page	string	Source page identifier
loaded_at	timestamp	Data ingestion timestamp

At this stage, data were minimally processed and retained in near-raw form to preserve original fidelity and enable backtracking if needed.

Silver Layer (Transformation and Integration)

The Silver Layer consolidates and standardizes the raw data into relational tables suitable for analytical queries and integration with external systems, which resulted in the creation of two tables.

Table 68: showtimes_dim_cinemas

Contains harmonized cinema-level metadata for geographical and organizational analysis.

Column	Type	Description
id	int	Cinema identifier
name	string	Cinema name
address	string	Cinema address
country	string	Country code
lat	double	Latitude

lon	double	Longitude
slug	string	URL-friendly cinema identifier
chain_id	int	Reference to cinema chain

Table 69: showtimes_fact_showings

Contains normalized screening-level data linking cinemas and movies

Column	Type	Description
id	string	Unique showing identifier
date	date	Screening date
cinema_id	int	Foreign key to showtimes_dim_cinemas
cinema_country	string	Country of cinema
is_id	int	Internal movie identifier
title	string	Movie title
tmdb_id	bigint	TMDB identifier
imdb_id	string	IMDB identifier

Gold Layer (Analytical Output)

The Gold layer aggregates curated data into analytical models that support descriptive and predictive analytics within FIDA. At this stage, each screening is uniquely identified by the tuple (country, cinema_id, movie_id, start_at).

The Gold layer integrates:

- Temporal aggregations (e.g., daily, weekly, and yearly screening counts);
- Market-level KPIs (e.g., screenings per cinema, screenings per title);
- Cross-database enrichment (e.g., linking with TMDB for genre and popularity metrics).

This layer powers analytical dashboards, recommendation engines, and modeling pipelines that examine audience behavior, screening density, and temporal exhibition trends across markets.

European Audiovisual Observatory Yearbook

Data collection and Processing

Data was retrieved for the EAO Yearbook Online Service, which was stored in an Excel table containing information about average ticket price, country ISO code, and year. Raw data was subsequently imported into the Bronze Layer of the Databricks data lake for further transformation.

Quality Control

While data retrieved was well organized, some Quality Control (QC) activities were performed in order to ensure interoperability.

1. Standardization of data.

- The field *country* was cross-checked against the aggregated country list in order to ensure compatibility and support interoperability with other tables.
- The data was reviewed in order to confirm it had a consistent number format

2. Consistency Check
    - The data across years within a single country was compared to detect anomalies or missing values.

Bronze Layer (Raw Data Ingestion)

Data was ingested into the Bronze layer of the Databricks environment in a separate volume (market). The data was stored in a table in Excel format, preserving the original field names and field conventions to ensure traceability and reproducibility.

Silver Layer (Transformation and Integration)

The information from the Bronze layer was transformed in order to create the following output:

- Crescine.silver.ticket_prices, which details the currency used, country (in ISO code), average ticket price by year, update timestamps and a rowhash.

Table 70. ticket_prices

Column	Type
Country	string
currency	string
Year_2014	double

Year_2015	double
Year_2016	double
Year_2017	double
Year_2018	double
Year_2019	double
Year_2020	double
Year_2021	double
Year_2022	double
Year_2023	double
Year_2023_to_2022	double
rowhash	string
source	string
loaded_at	timestamp
updated_at	timestamp
file	string

Gold Layer (Analytical Output)

The Gold layer integrates the EAO Yearbook data as a dimension table thus allowing:

- Analyzing the relationships between admissions, box-office and ticket prices.
- Enriching the predictions made based on historical market data.

Therefore, this information helps to deepen the study of theatrical markets within Europe and provides context related to admissions and box-office data.

Linked References